Sangram Nandi

Software Engineer

sangramnandi99@gmail.com 9875369598 Kolkata, India linkedin.com/in/sangram-nandi-2831ba180

sangram-sogftware-engineer.vercel.app

SUMMARY

Software Engineer with **nearly 4 years of experience** in **Full Stack Development and Cross-Platform Mobile Application Development**, specializing in **React.js, React Native, JavaScript (ES6+), TypeScript, Next.js, Node.js, Express.js, MongoDB, Redux, Context API, Material UI, HTML5, CSS3, and RESTful APIs**.Experienced in building **scalable web applications, cross-platform mobile applications, backend services, and RESTful APIs**, delivering seamless experiences across **web, Android, and iOS platforms**. Proficient in **frontend and backend architecture, state management, database integration, authentication, reusable component development, API integration, and application performance optimization**, with a strong focus on developing **secure, maintainable, high-performing, and user-centric software solutions**.

PROFESSIONAL EXPERIENCE

Software Engineer 11/2022 – 08/2026
Metavy OPC Private Limited Kolkata, India

- **Engineered scalable web and cross-platform mobile applications** using **React.js, React Native, Next.js, JavaScript (ES6+), TypeScript, Redux, Context API, HTML5, CSS3, and Tailwind CSS**, improving code maintainability by **30%** through modular architecture and reusable components.
- **Developed responsive web interfaces** using **React.js, Next.js, Tailwind CSS, and Material UI**, translating UI/UX designs into intuitive experiences and maintaining consistent layouts across browsers and mobile devices, contributing to a **25% improvement in usability**.
- **Built Android and iOS applications** using **React Native**, implementing adaptive layouts, platform-specific functionality, reusable components, and responsive design principles for consistent cross-platform experiences.
- **Developed backend services and RESTful APIs** using **Node.js and Express.js**, implementing server-side business logic, API endpoints, middleware, request validation, and application workflows.
- **Integrated MongoDB** for structured data storage, retrieval, and management, supporting efficient database operations and reliable communication between application layers.
- **Integrated RESTful APIs** across web and mobile clients, managing asynchronous requests, response handling, loading states, exception management, and data synchronization to reduce API-related failures by **35%** and improve data reliability by **30%**.
- **Implemented centralized state management** using **Redux and Context API**, establishing predictable data flows and reducing state-related issues by **25%**.
- **Implemented secure authentication and authorization** using **JWT, access and refresh tokens, protected routes, session persistence, role-based access control, and email OTP verification**, improving authentication reliability by **30%**.
- **Integrated Google OAuth 2.0** for social authentication across web and mobile applications, simplifying account access and reducing login friction by **25%**.
- **Integrated SendGrid and Nodemailer** for transactional emails, OTP verification, automated notifications, and account-related communication workflows.
- **Optimized application performance** through lazy loading, code splitting, memoization, optimized rendering, and React Native FlatList tuning, reducing page load time by **40%** and improving responsiveness by **30%**.
- **Strengthened application reliability** using TypeScript interfaces, generics, shared structures, and reusable utilities, while developing **5+ custom hooks and component libraries** to reduce runtime defects and code duplication by **25% and 35% respectively**.
- **Optimized large data collections** using **React Native FlatList**, implementing pagination, unique keys, reusable list items, and controlled updates for efficient memory utilization and smooth scrolling.
- **Implemented form validation and application safeguards** through controlled inputs, input sanitization, structured error handling, and client-side validation, improving data accuracy by **30%**.
- **Integrated Cloudflare Turnstile** for bot protection and user verification, implementing token generation and backend validation to mitigate automated and malicious requests.
- **Monitored production applications using Sentry**, collaborating with QA teams to investigate defects, validate fixes, and perform root-cause analysis, contributing to a **20% reduction in post-release issues**.
- **Collaborated with Product Owners, UI/UX Designers, Backend Engineers, and QA professionals** throughout the SDLC, delivering **15+ business-critical features** with a **95% on-time sprint completion rate**.

- **Managed Git-based development workflows** through feature branches, structured commits, merges, and collaborative code reviews, reducing integration conflicts by **20%**.

EDUCATION

Master of Computer Applications (MCA) 07/2020 – 08/2022
Guru Nanak Institute of Technology | CGPA: 9.2/10 West Bengal, India

Bachelor of Computer Applications (BCA) 06/2017 – 06/2020
Techno India College of Technology | CGPA: 7.1/10 West Bengal, India

SKILLS

Programming Languages:
JavaScript (ES6+), TypeScript

State Management
Redux, Context API

Database
MongoDB

Mobile Development
React Native, Cross-Platform Development, Android, iOS

Third-Party Integrations
SendGrid, Nodemailer

Frontend
React.js, Next.js, HTML5, CSS3, Tailwind CSS, Material UI

Backend
Node.js, Express.js, RESTful APIs

Authentication & Security
JWT, Access & Refresh Tokens, Google OAuth 2.0, Role-Based Access Control, Cloudflare Turnstile, Express Rate Limit

Development Tools
Git, GitHub, Vite, NPM, Postman, VS Code, Sentry

CERTIFICATES

- Introduction to Generative AI — Simplilearn (Google Cloud)
- Frontend Developer (React) — HackerRank
- MERN Stack Fundamentals — Great Learning
- AI for Everyone — Coursera

PROJECTS

Online Car E-Auction Platform — SaaS Application
Technologies: React.js, React Native, Node.js, Express.js, MongoDB, RESTful APIs, JWT, SendGrid
- Developed a **SaaS-based online car auction platform** with web and cross-platform mobile application experiences, enabling showroom owners to list and manage vehicle auctions while customers can securely browse vehicles and participate in bidding.

Cross-Platform Digital Wallet
Technologies: React.js, React Native, Redux, RESTful APIs
- Developed a **digital wallet application** with web and cross-platform mobile interfaces, supporting account management, authentication, financial operations, and protected data handling.

Transaction Monitoring System
Technologies: React.js, React Native, Redux, JavaScript, RESTful APIs
- Developed a transaction monitoring solution with web and mobile application interfaces for tracking account activities, transaction records, and historical financial operations.

Digital Identity & Wallet Application
Technologies: React.js, React Native, Redux, JavaScript, TypeScript ethers.js, BIP39, Web3 services
- Developed a digital identity and wallet application with web and cross-platform mobile experiences, focusing on account onboarding, digital address generation, identity verification, and account recovery.

Financial Trading Platform
Technologies: React.js, Redux, Node.js, Express.js, MongoDB, RESTful APIs, JWT, SendGrid
- Developed a financial trading platform with web, supporting account management, wallet operations, secure transactions, portfolio monitoring, and transaction history.

AWARDS

JIS Innovation Award 03/2022
Recognized for **research and innovation in Deep Learning and Neural Networks**, demonstrating research-oriented problem-solving and exploration of neural network technologies.